

Scale-Specific Viscoelastic Characterization of Hydrogels: Integrated AFM and Finite Element Modeling

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Abstract

Viscoelastic hydrogels mimic the dynamic mechanical properties of native extracellular matrices, making them essential for biomedical applications. However, characterizing their scale-dependent mechanical properties remains challenging, despite their critical influence on cell-material interactions and biomaterial performance. Here, we present an integrated experimental-computational approach to quantify and model the viscoelastic behavior of interpenetrating polymer network hydrogels across micro- and macro-scales. Atomic force microscopy-based stress relaxation tests revealed that microgels exhibit rapid, localized relaxation, while macroscopic bulk gels displayed prolonged relaxation dominated by poroelastic effects. Finite element simulations accurately replicated experimental conditions, enabling the extraction of key parameters: instantaneous elastic modulus, relaxation modulus, and relaxation time constant. We further developed a novel analytical model to predict viscoelastic parameters from experimental data with minimal error (< 5%), significantly streamlining characterization. Our findings highlight the necessity of scale-specific mechanical analysis and provide a robust platform for designing biomaterials with tailored viscoelasticity for tissue engineering and regenerative medicine.

Keywords

viscoelastic hydrogels, interpenetrating polymer networks, microgels, bulk hydrogels, scale-dependent mechanics, atomic force microscopy, stress relaxation, finite element modeling

Introduction

Hydrogels have transformed biomedical engineering by mimicking the dynamic, water-rich environment of native extracellular matrices (ECM) through tunable, three-dimensional polymer networks derived from natural or synthetic sources [1–3]. These materials replicate the porous architecture and biochemical signaling of tissues, enabling physiologically relevant models that surpass traditional two-dimensional cultures. As such, hydrogels are integral to advances in targeted drug delivery [4–7], cell encapsulation [8,9], tissue engineering [10–13], biosensing, and mechanobiology [14–18].

A key advantage of hydrogels is their dual tunability in both chemical composition and structural scale. Bulk hydrogels typically serve as robust scaffolds for tissue regeneration, promoting cell adhesion, proliferation, and differentiation, while enabling advanced applications such as 3D bioprinting and organ-on-a-chip platforms [19–24]. In contrast, microscale hydrogel particles, known as microgels (ranging from 0.1 to 100 μm in diameter), excel in cell encapsulation, controlled drug release, and biosensing [25,26], with recent studies demonstrating their ability to guide endothelial morphogenesis through electrostatic microenvironment modulation [27].

Despite their growing utility, microgel mechanics are often extrapolated from bulk gel measurements, assuming that identical formulations yield consistent behavior across scales. However, significant discrepancies exist between micro- and macroscale properties of the same material [28,29]. These differences critically impact biological performance, microgel stiffness and stress relaxation directly affecting cell proliferation, differentiation, and mechanotransduction [30–35]. Inaccurate characterization thus risks compromising biomedical outcomes.

Recent advances in hydrogel technology focus on viscoelastic systems that more accurately recapitulate the time-dependent mechanical behavior of living tissues

[36–40]. Unlike purely elastic materials, viscoelastic hydrogels resist immediate deformation and dissipate energy under load, mimicking ECM remodeling during physiological processes such as wound healing [41–43]. However, characterizing these dynamics – particularly at microscale – remains challenging. While bulk rheometry captures macroscale viscoelasticity, probing microgels demands specialized techniques such as atomic force microscopy (AFM)–based nanoindentation [44,45]. Yet, standard stress-relaxation tests under constant strain often oversimplify the complex time-dependent soft material responses, leading to inaccurate estimates of relaxation time and creep compliance.

To address these challenges, we employ previously established viscoelastic interpenetrating polymer network (IPN) hydrogels [46] as model systems to investigate correlations in viscoelastic parameters across multiple scales. These hydrogels feature a dual-network architecture: (1) a covalent network formed through Michael-type addition between thiolated four-armed poly(ethylene glycol) (starPEG) and maleimide-functionalized sulfated glycosaminoglycan heparin, providing structural integrity and elastic behavior; and (2) a physical network arising from reversible electrostatic interactions between heparin and heparin-binding peptides conjugated to starPEG, imparting a time-dependent viscous response (**Figure 1A**) [47,48]. This design allows precise control over both stiffness and stress relaxation behavior, making these IPN hydrogels ideal for exploring scale-dependent viscoelastic phenomena.

By integrating AFM-based stress relaxation tests with finite element (FE) simulations, we dissected scale dependent viscoelasticity, revealing that bulk gels exhibit prolonged poroelastic relaxation – where water migration through an extensive pore network contributes to a viscous response [49–52], while microgels display rapid, localized relaxation. These scale-dependent differences underscore that inferring microgel

properties from bulk gel measurements can lead to substantial discrepancies in key parameters such as relaxation time and creep compliance.

We further developed an analytical model to predict the viscoelastic properties of microgels with error margins below 5%, streamlining characterization. This approach directly correlates AFM measurements with optimized material parameters, offering unprecedented accuracy in capturing both elastic and viscous contributions to hydrogel behavior. Ultimately, our approach bridges micro-macro mechanical discrepancies, offering critical insights for biomaterial design.

Results and Discussion

Formation of Viscoelastic Microgels and Bulk Hydrogels

Viscoelastic microgels and bulk hydrogels were fabricated using an established cell-instructive interpenetrating polymer network (IPN) hydrogel platform (**Figure 1A**) [46]. This system combines two distinct networks: (1) a covalent network formed through Michael-type addition between thiolated star-shaped polyethylene glycol (starPEG) and maleimide-functionalized heparin that provides elastic properties; and (2) a physical network of reversible electrostatic interactions between heparin-binding peptides ((KA)₇) conjugated to starPEG and negatively charged heparin sulfate groups, imparting viscous properties and enabling controlled stress relaxation. By adjusting the ratio of these two networks, the viscoelastic profile of the hydrogel can be precisely tuned to mimic natural tissue mechanics.

For the first time, monodisperse microgels were fabricated from this IPN system using microfluidic techniques with integrated electrodes, enabling rapid and controlled mixing of hydrogel precursors via emulsion droplet coalescence (**Figure 1B**) [53]. Droplets of the PEG-based precursor solution (DP 1) were generated at a flow-focusing junction using a fluorinated oil continuous phase (CP) stabilized by surfactant. These droplets

were then guided to merge with heparin-based precursor droplets (DP-2) at a T-junction, where an applied electric field (50 kV) triggered rapid coalescence by destabilizing the water-oil interface. Subsequent mixing in a meandering channel facilitated uniform gelation, producing highly monodisperse microgels (diameter = $67.8 \pm 2.4 \mu\text{m}$, **Figure 1C**).

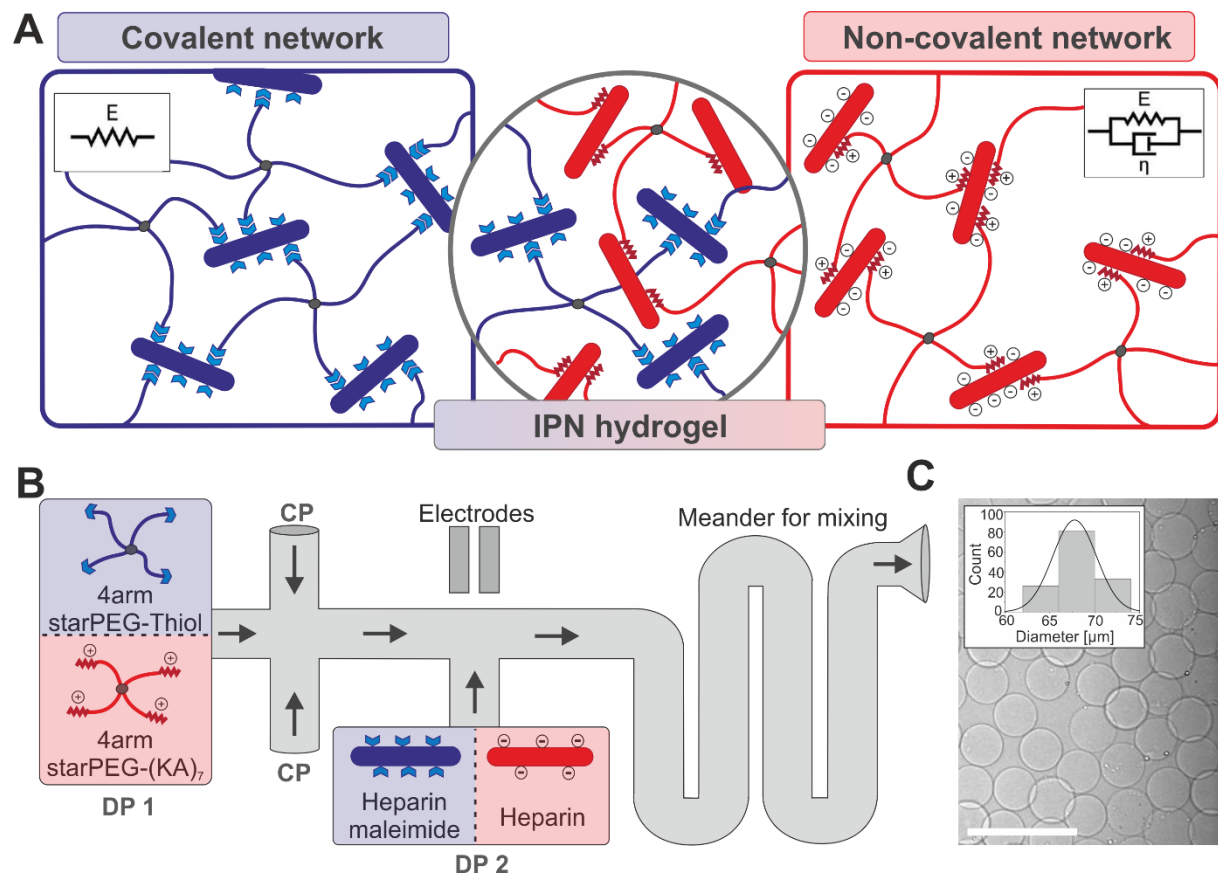


Figure 1: Composition and fabrication of viscoelastic IPN microgels. (A) Schematic representation of the viscoelastic interpenetrating polymer network (IPN) hydrogel platform. The hydrogel consists of two networks: a covalent network (blue) formed by a Michael-type addition reaction between thiolated starPEG and maleimide-functionalized heparin provides robust elastic properties. Concurrently, a physically crosslinked network (red) formed via reversible electrostatic interactions between sulfated heparin and starPEG conjugated with the heparin-binding peptide (KA)₇ imparts viscous properties and controlled stress relaxation [46]. (B) Microfluidic fabrication of IPN microgels [53]. A microfluidic device equipped with integrated electrodes enables rapid and precise mixing of the continuous phase (CP) of fluorinated oil to which a surfactant is added for droplet stabilization and the precursor solutions. Droplets containing PEG-based precursors (dispersed phase 1 – DP1) and heparin-based precursors (dispersed phase 2 – DP2) are generated separately, merged at a T-junction through electric field-induced droplet fusion, and uniformly mixed within a downstream meandering channel to form monodisperse microgels. Arrows indicate fluid flow directions. (C) Brightfield microscopy image of the resulting viscoelastic IPN microgels. The inset shows the size distribution of the microgels (mean = $67.8 \pm 2.4 \mu\text{m}$, $n = 140$). Scale bar: $250 \mu\text{m}$.

Mechanical Characterization and FE Simulations Reveal Scale-Dependent Behavior

To investigate the scale-dependent mechanical properties of the IPN hydrogel, we conducted AFM-based nanoindentation experiments on both microgels and bulk hydrogels prepared from identical precursor compositions. In these experiments, a probe of defined geometry indents the sample while the cantilever deflection measures the applied force. Fitting the resulting force–indentation curves to the Hertz model yielded the Young’s modulus (E_0). Our results revealed that microgels are significantly more compliant than bulk gels, and both formats stiffen with increasing indentation velocity, indicating pronounced time-dependent viscoelastic behavior (**Figure 2A**).

To further characterize the viscoelastic properties, we performed stress relaxation tests at multiple indentation velocities, following a three-phase protocol (**Figure 2B**). In the first phase, the AFM cantilever approached the sample surface at constant velocity until reaching a predetermined indentation force, inducing instantaneous elastic deformation. Continuous recording of cantilever deflection during this phase provided force–indentation data directly correlated to the material’s instantaneous elastic modulus. These initial measurements serve as the baseline for understanding the load-bearing capabilities of the sample. In the second phase, the cantilever position was held constant at maximum indentation for 10 s, during which the initial stress decays as the polymer network rearranges. This phase yields key viscoelastic parameters – relaxation modulus (E_I) and time constant (τ) – while the residual stress isolates the purely elastic component (**inset, Figure 2B**). In the third phase, the cantilever was retracted from the sample surface. However, this phase was omitted from quantitative analysis due to adhesion artifacts that obscure intrinsic material behavior.

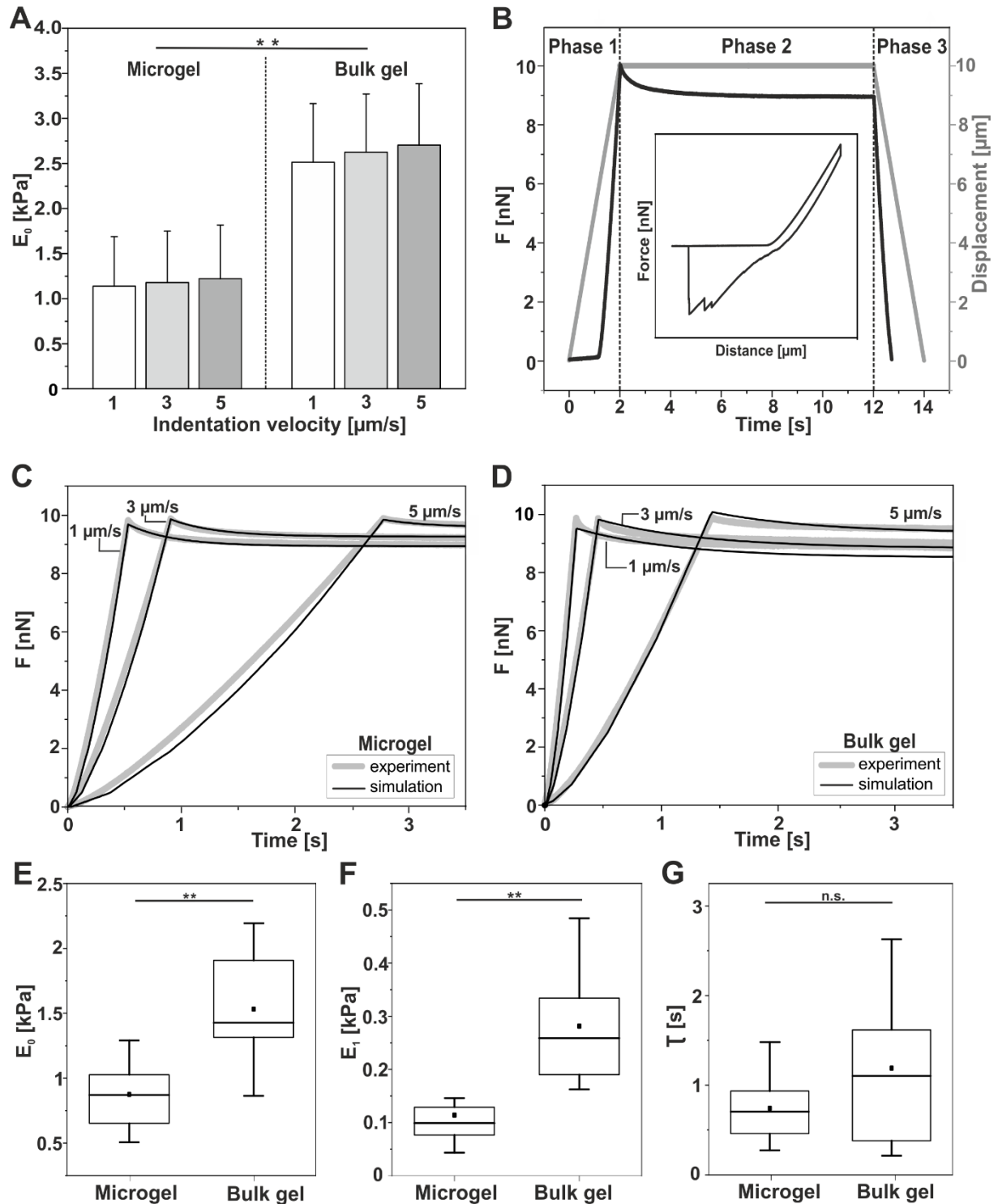


Figure 2: Comparative mechanical characterization of viscoelastic microgels and bulk hydrogels. (A) AFM-based nanoindentation analysis comparing elasticity (Young's modulus, E_0) of microgels and bulk gels at varying indentation velocities. Data are presented as mean $\pm$ s.d. ($n=10$). (B) Schematic illustration of the three phase AFM-based stress relaxation protocol: Phase 1 (Indentation): The cantilever approaches and indents the sample at constant velocity until a predefined contact force (10 nN) is reached. Phase 2 (Stress Relaxation): The cantilever position is held constant (for 10 s) at maximum displacement, to monitor force decay due to material relaxation. Phase 3 (Retraction): The cantilever is withdrawn from the sample surface.

The graph shows representative cantilever height (grey) and contact force (black) profiles over time, with inset illustrating the force–height relationship. **(C, D)** Experimental (grey) and simulated (black) relaxation curves for microgels **(C)** and bulk gels **(D)** across indentation velocities. **(E–G)** FE-derived viscoelastic parameters. Box plots show 25th/75th percentiles (boxes), means (squares), medians (lines), and standard deviation (whiskers). ****P < 0.01** (Wilcoxon–Mann–Whitney test).

Both microgels (**Figure 2C**) and bulk gels (**Figure 2D**) demonstrated distinct velocity-dependent relaxation profiles characteristic of viscoelastic materials. To quantitatively extract viscoelastic parameters from these curves, FE simulations replicating the AFM experiments were performed (see Methods). The simulated stress relaxation curves (black lines, **Figure 2C and D**) closely matched experimental data across different indentation velocities, validating the measurement approach and confirming the observed velocity dependence.

However, bulk gels displayed significant deviations during the relaxation phase (Phase 2), particularly at higher indentation velocities, characterized by prolonged and inconsistent relaxation patterns (**Supplementary Figure 1**). This discrepancy likely arises from two competing relaxation mechanisms: (1) intrinsic viscoelasticity arising from polymer network rearrangements, and (2) poroelastic effects caused by fluid migration through the gel's porous structure during sustained deformation [49,50,54]. The FE simulations, which modeled only the viscoelastic response without accounting for poroelastic effects, consequently showed discrepancies with experimental data during extended relaxation periods.

Bulk gels exhibit prolonged relaxation effects beyond ten seconds, primarily due to poroelastic fluid migration through their extensive pore networks [55,56], as evidenced by non-converging contact forces during indentation. In contrast, microgels exhibited primarily localized viscoelastic relaxation with minimal poroelastic contribution, likely due to their smaller dimensions restricting fluid movement. These differences likely

arise from both dimensional effects and fabrication-induced structural variations: bulk gels are formed via active mixing, while microgels rely on diffusion-driven mixing in microfluidic channels, potentially introducing subtle structural variations.

Direct comparison of the FE-derived mechanical parameters revealed significant scale-dependent differences between microgels and bulk hydrogels, despite identical hydrogel composition. Specifically, microgels showed significantly lower Young's moduli (E_0) compared to bulk gels (**Figure 2E**), along with substantially reduced relaxation moduli (E_1 ; **Figure 2F**) and shorter relaxation times (τ ; **Figure 2G**). These findings demonstrate that hydrogel mechanical properties cannot be simply extrapolated across length scales, highlighting the critical importance of scale-specific mechanical characterization for accurate biomaterial design and application.

Characterization of Material Parameters Using Predefined Equations

While the described optimization procedure accurately determines viscoelastic material parameters, it becomes inefficient when applied to large experimental datasets. To address this limitation, we developed a streamlined analytical approach based on a set of predefined equations that extract key viscoelastic parameters from just four experimental measurements (**Figure 3A**). In addition to these four measurements, the microgel radius (R) and the maximum indentation force (F_1 , marking the transition between phases 1 and 2) are required. The governing equations are:

$$E_0 = \alpha_1 \frac{F_0}{u_{max} \alpha_2 R^{\alpha_3}},$$

$$\tau = \beta_1 (t_{mid} - t_1),$$

$$E_1 = \gamma_1 \left[\frac{F_1 - F_0}{F_1} \right]^{\gamma_2} e^{\gamma_3 \frac{t_1}{\tau}} E_0,$$

where F_1 is the maximum contact force between cantilever and microgel, F_0 is the residual contact force at the end of phase 2, t_1 and t_{mid} denote the start and midpoint times of phase 2 (with $F_{mid} = (F_1 + F_0)/2$), and u_{max} is the maximum indentation depth during phase 2 (**Figure 3B**). The coefficients ($\alpha_1, \alpha_2, \alpha_3, \beta_1, \gamma_1, \gamma_2, \gamma_3$) were optimized across experimentally relevant ranges of microgel radii, maximum contact forces, cantilever velocities, and material parameters (E_0 , E_1/E_0 and τ , see Methods).

Using these optimized coefficients, the predefined equations achieved maximum relative errors of 3.1% for E_0 , 1.3% for τ and 6.0% for E_1 across the investigated parameter space. Detailed error analyses (**Figure 3C-F**) illustrate the accuracy of these equations for microgel radii ranging from 20 μm to 50 μm , maximum contact forces from 2 nN to 10 nN, and Young's moduli up to 2000 Pa. Consistent with experimental observations, the relaxation modulus (E_1) was considered up to 20% of the Young's modulus (E_0). Specifically, **Figure 3C** shows the relative error in calculated E_0 ($E_{0,cal}$) across varying F_1 and R , with the absolute error peaking at $E_0 = 2000$ Pa and $F_1 = 2$ nN. Similarly, the relative error for the relaxation time constant τ remained consistently low (1.3%, **Figure 3D**). The absolute error in calculating E_1 was most pronounced at Young's Moduli (E_0) of 500 Pa and 2000 Pa, particularly when the difference between F_1 and F_0 was minimal, corresponding to the lowest viscosity condition. **Figures 3E and 3F** illustrate these absolute errors in the relaxation modulus (E_1) at $E_0 = 500$ Pa and $E_0 = 2000$ Pa, respectively, at a cantilever velocity $v = 3 \mu\text{m/s}$ and $\tau = 0.5$ s, varying F_1 and microgel radii (R). Comparable absolute errors for E_1 were observed at intermediate Young's moduli ($E_0 = 1000$ Pa and 1500 Pa).

Notably, at very low cantilever velocities (i.e. 1 $\mu\text{m/s}$), the apparent viscosity decreases significantly, causing the predefined equations for E_1 to exhibit higher relative errors (up to 30%). Additionally, the equations were specifically optimized for a spherical AFM probe radius of 5 μm . For significantly smaller microgels ($< 20 \mu\text{m}$) or different

cantilever geometries, re-optimization of the coefficients α , β , γ may be necessary. Within the validated parameter ranges, however, these equations provide a rapid, accurate, and straightforward method for extracting viscoelastic material parameters from AFM-based stress relaxation experiments, significantly enhancing evaluation efficiency without significant compromises in accuracy.

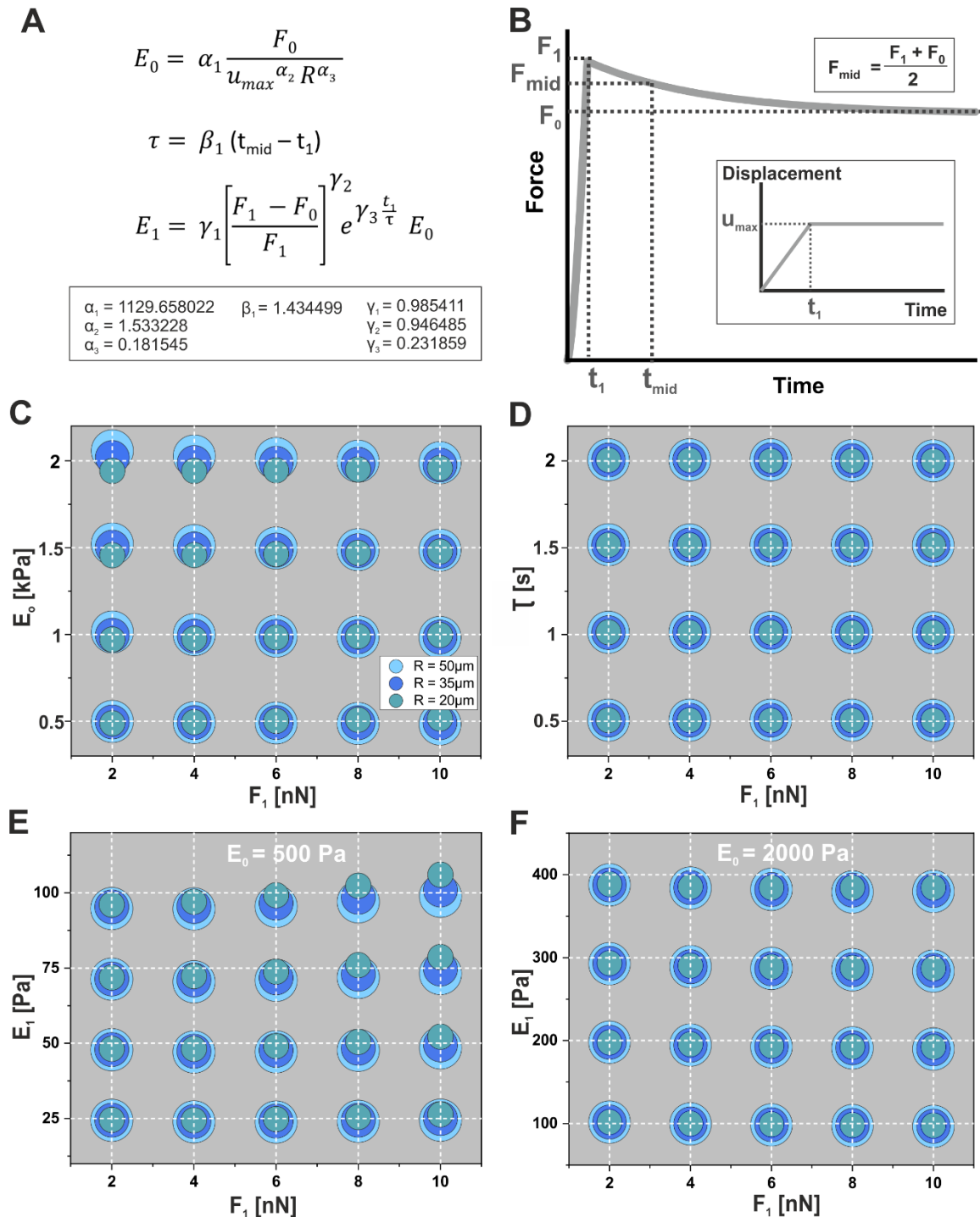


Figure 3: Determination of viscoelastic material parameters and associated errors: (A) Predefined equations enable fast estimation of key viscoelastic parameters – fully relaxed Young’s modulus (E_0), relaxation modulus (E_1), and relaxation time constant (τ) – directly from AFM-based stress relaxation data. **(B)** Representative force–time and indentation–time curves illustrating the experimental input parameters required for the calculations shown in (A). **(C–F)** Error analysis comparing estimated parameter values (dots) to actual values (grid intersections) as functions of the maximum contact force (F_1) and the microgel radius (R): **(C)** Maximum relative errors for Young’s modulus (E_0). **(D)** Maximum relative errors for relaxation time (τ), observed under conditions of highest relaxation (cantilever velocity $v = 5 \mu\text{m/s}$, $E_0 = 2000 \text{ Pa}$, $E_1 = 400 \text{ Pa}$). **(E, F)** Maximum relative errors for relaxation modulus (E_1) at the lower ($E_0 = 500 \text{ Pa}$, **E**) and upper ($E_0 = 2000 \text{ Pa}$, **F**) limits of tested Young’s moduli under conditions of lowest relaxation.

Conclusion

This study introduces an integrated approach for characterizing scale-dependent viscoelastic properties in hydrogels, combining advanced microfluidic fabrication, AFM-based nanoindentation, and computational modeling.

We employed a modular hydrogel system composed of two IPNs, enabling precise and independent control over elasticity and stress relaxation. For the first time, monodisperse microgels were fabricated from this IPN system using microfluidic techniques with integrated electrodes, enabling rapid and controlled mixing of hydrogel precursors via emulsion droplet coalescence. Bulk gels prepared from identical formulations provided a direct basis for comparing viscoelastic properties across different length scales. AFM nanoindentation experiments, supported by FE simulations, revealed that hydrogel mechanics cannot be extrapolated across length scales, highlighting the necessity of scale-specific characterization for accurate biomaterial design.

To facilitate practical implementation, we developed a universal analytical framework that rapidly extracts key viscoelastic parameters – immediate elastic modulus (E_0), relaxation modulus (E_1), and relaxation time (τ) – from AFM stress relaxation data. This approach achieves remarkable accuracy (maximum errors: 3.1% for E_0 , 1.3% for τ ,

6.0% for E_1) across physiologically relevant ranges while eliminating the need for computationally intensive optimization.

This integrated experimental-computational strategy provides a robust platform for designing viscoelastic hydrogels with tailored properties and is particularly valuable for applications in tissue engineering and cellular mechanobiology, where precise control over time-dependent mechanical properties is crucial for guiding cell behavior.

Materials and Methods

Hydrogel Components

The IPN hydrogels were prepared from starPEG and heparin precursors. The starPEG components included a commercially sourced starPEG-Thiol (0.83 mM; MW 10,600 g/mol; Polymer Source, Inc.) and an in-house synthesized starPEG-(KA)₇ (0.28 mM; MW 17,000 g/mol). The heparin components comprised non-functionalized heparin (0.28 mM; MW 14,000 g/mol; Merck KGaA) and an in-house synthesized maleimide-functionalized heparin HM6 (0.83 mM; MW 15,000 g/mol). The viscoelastic properties of the IPN hydrogels were precisely tuned by controlling the network ratio, with the physical (non-covalent) network constituting 25% of the total molar composition and the covalent network comprising the remaining 75%.

Microfluidic Device Fabrication and Setup Design

Water-in-oil emulsions were generated using custom poly (dimethylsiloxane) (PDMS) microfluidic devices fabricated through standard photolithography and soft lithography techniques [53]. The microfluidic channels were uniformly designed with a width of 50 μm for the dispersed phases, continuous phase, and droplet-transporting channel. The fabrication process began by spin-coating a 3-inch silicon wafer (Siegert Wafer)

with SU-8 2025 photoresist (Micro Resist Technology) to create a master mold. Channel patterns were defined using a mask aligner (MJB3, Suess MicroTec) and a printed film mask. After exposure and baking, the uncured photoresist was removed using a developer (mr-Dev 600, Micro Resist Technology). The microchannel structure was then replicated via PDMS replica molding. A mixture of PDMS base and curing agent (Sylgard® 184, Dow Corning) at a 10:1 ratio was degassed using a planetary centrifugal mixer (ARE-250, Thinky), poured over the master mold and cured at 65°C for 2 hours. After demolding, 1.0 mm inlet/outlet ports were punched with a biopsy tool (KAI Medical), and the PDMS channel side was bonded to a glass microscopy slide via oxygen plasma treatment (80W for 15 s, MiniFlecto 10, Plasma Technology). To enable electric field-assisted droplet manipulation, electrode microchannels were filled with a low-melting metal alloy (51% In, 32.5% Bi, 16.5% Sn, Indium Corporation of America) to generate localized strong electric fields. Non-electrode microchannels were hydrophobized by injecting a 1% (v/v) solution of (tridecafluoro-1,1,2,2-tetrahydrooctyl)trichlorosilane (Gelest) in Novec 7500 (IoLiTec). During operation, dispersed phases were delivered through 500 μ L gastight syringes (Hamilton) and continuous phases via 3 mL disposable syringes (BD Luer-lock), all controlled by high-precision syringe pumps (Pico Plus Elite) connected through PE tubing (0.38 mm ID). The emulsion process was monitored in real-time using an inverted bright-field microscope (Axio Vert.A1, Carl Zeiss) equipped with a high-speed digital camera (Miro C110, Vision Research Inc.).

Microgel Fabrication

Microgel fabrication was initiated by injecting the first dispersed phase (DP-1), containing 0.83 mM starPEG-Thiol and 0.28 mM starPEG-(KA)₇, at a flow rate of 50 μ L hr⁻¹ (**Figure 1B**). Emulsification was achieved using two continuous phases of 2%

(w/w) PFPE-PEG-PFPE triblock copolymer surfactant (RAN Biotechnologies) in Novec 7500, delivered at 500 $\mu\text{L hr}^{-1}$. The second dispersed phase (DP-2), containing 0.83 mM heparin-HM6 and 0.28 mM heparin, was introduced through a side channel at matching flow rates, creating paired DP-1/DP-2 droplets. These pairs coalesced into single droplets upon exposure to a 50 kV electric field at integrated electrodes. Subsequent rapid mixing and gelation occurred in a meandering channel, driven by the chemical reaction between the components of DP-1 and DP-2. Microgels were collected every 10 minutes and purified through three washes with a 20% (v/v) 1H,1H,2H,2H-perfluoro-1-octanol (PFO, Sigma-Aldrich) in Novec 7500, before being transferred to PBS buffer.

Bulk Gel Preparation

Preparation of IPN bulk gels involved two separate precursor solutions: (1) a starPEG solution prepared by combining starPEG-(KA)₇ with starPEG-Thiol, and (2) a heparin solution prepared by combining non-functionalized heparin with heparin HM6. These solutions were mixed at a 25:75 molar ratio of non-covalent to covalent networks – matching the microgel formulation – and deposited as droplets (20 μL) in Petri dishes filled with PBS. After crosslinking, the resulting hydrogel droplets measured approximately 5 mm in diameter and 1 mm in height.

Mechanical Characterization of Microgels and Bulk Gels

Mechanical properties were quantified using a NanoWizard 4 AFM (Bruker) coupled with an inverted optical microscope (Observer A1, Zeiss). A tipless cantilever (PNP-TR-TL, NanoWorld; $k = 0.08 \text{ N/m}$) fitted with a 10 μm colloidal force probe (Microparticles GmbH) was employed for both indentation and stress relaxation measurements. The cantilever was calibrated before each experiment using the

thermal noise method [57] with all measurements conducted in PBS at room temperature.

For indentation testing, the colloidal probe was positioned over the center of individual microgels or bulk gels. Force-distance curves were recorded at three approach/retract velocities (1, 3 and 1 $\mu\text{m/s}$). A constant contact force of 2 nN was maintained, resulting in typical indentation depths of approximately 1 μm . The Young's modulus was determined by fitting the approach segment of the force–distance curve to the Hertz model for a spherical indenter. Microgels analysis incorporated a double-contact correction to account for additional deformation from the counter pressure at the microgel's underside [58–60]. All data processing was performed using the AFM's native software with custom analysis algorithms.

Stress relaxation measurements followed a three-step protocol (**Figure 2B**). In the indentation phase (Phase 1), the cantilever advanced at controlled velocities (1, 3, or 5 $\mu\text{m/s}$) until reaching a 10 nN loading force. The subsequent hold phase (Phase 2) maintained constant displacement for 10 seconds while monitoring force decay, allowing characterization of time-dependent viscoelastic behavior. Finally, in the retraction phase (Phase 3), the cantilever was withdrawn from the sample surface.

Determination of FE Models

To computationally replicate the AFM experiments, FE models were developed representing both microgel and bulk gel geometries. Microgels were modelled as spheres (**Supplementary Figure 2A**), while bulk gels were approximated as cylinders with 1000 μm height to match experimental conditions (**Supplementary Figure 2B**). Both geometries incorporated rotational symmetry to optimize computational efficiency.

A mesh convergence study established the optimal element density for accurate parameter determination (**Supplementary Figure 2C**). The microgel model comprised 23,690 elements (24,130 degrees of freedom), while the bulk gel model used 14,714 elements (15,080 degrees of freedom). Triangular elements with quadratic shape functions were employed throughout, with localized mesh refinement in the bulk gel's contact region to resolve steep stress gradients. The AFM indenter was modeled as a rigid 10 μm sphere with frictionless contact assumptions.

Finite Element Simulation-Based Analysis of Viscoelastic Material Properties

FE simulations were employed to analyze the viscoelastic behavior of microgels and bulk gels, coupled with an optimization procedure to determine material parameters that best fit experimental data. Given the significant displacements observed during AFM indentation, the material model was based on finite strain theory. The viscoelastic material behavior was described using a combination of a Neo-Hookean model and a single-term Prony series. The Neo-Hookean model captured the nonlinear elastic response at large deformations, while the Prony series component accounted for the time-dependent viscous response. This approach is equivalent to a generalized Maxwell model with a single spring-dashpot element, commonly used in relaxation experiments [61–63].

The time-dependent Young's modulus $E(t)$ is expressed in one dimension as:

$$E(t) = E_0 + E_1 e^{-(t/\tau)},$$

where E_0 is the fully relaxed Young's modulus, E_1 represents the initial viscous contribution, and τ is the relaxation time constant. The instantaneous (unrelaxed) modulus is given by $E_u = E_0 + E_1$, and τ can be interpreted as $\tau = \eta/E_1$, with η being the dashpot viscosity. To account for the nearly incompressible nature of the hydrogels, the Poisson's ratio was set to $\nu = 0.45$ [14,59].

Parameter optimization proceeded in two sequential phases:

1. Relaxed Modulus (E_0): The fully relaxed behavior was determined using phase 2 of the experiment, during which the indenter was held at the maximum displacement (u_{max}) for 10 seconds (**Figure 2B**).
2. Viscous Parameters (E_1 and τ): These parameters were optimized using the stress relaxation data from phase 2, where the displacement remained constant, and the stress decay was governed solely by viscous behavior.

This two-step approach minimized inaccuracies caused by potential nonlinear elastic effects not fully captured by the Neo-Hookean model during the initial loading phase.

FE Simulation and Parameter Optimization

FE simulations were performed in Abaqus (Dassault Systèmes. Abaqus 2024: Unified FEA Software), a commercial finite element software, with optimization implemented in Python (Python Software Foundation (2023)) using the “mystic” library, which combines evolutionary strategies with gradient-based methods.

Elastic Modulus (E_0) Optimization

E_0 was optimized using the maximum indenter displacement (u_{max}) from the experiments as the boundary condition. Since experiments were conducted at three indenter velocities ($v_1 = 1 \mu\text{m/s}$, $v_2 = 3 \mu\text{m/s}$, and $v_3 = 5 \mu\text{m/s}$), three separate simulations were executed for each candidate E_0 value. In these simulations, the material was modeled using a Neo-Hookean model (without viscosity). The simulated contact force (F_{sim}) at the end of the relaxation phase (phase 2) was compared with the corresponding experimental force (F_{exp}). The first objective function z_1 of the optimization was defined as:

$$z_1 = \sum_{i=1}^3 \sqrt{(F_{\text{sim},i} - F_{\text{exp},i})},$$

where $F_{\text{sim},i}$ and $F_{\text{exp},i}$ are the contact forces at the end of phase 2 for the three velocities (v_1 , v_2 and v_3).

Optimizing the Viscous Parameters (E_1 and τ)

The viscous parameters (E_1 and τ) were optimized by capturing the full force–time response during phases 1 and 2. In these simulations, the experimental displacement–time profile was imposed as the boundary condition. The simulated force–time curves (F_{sim}) were compared with the experimental curves (F_{exp}) at all time steps across the three velocities. The second objective function (z_2) was defined as:

$$z_2 = \sum_{i=1}^3 \sqrt{\sum_{j=1}^{n_i} \left(\frac{(F_{\text{sim},i,j} - F_{\text{exp},i,j}) t_j}{t_{\text{ges}}} \right)^2},$$

where i corresponds to the different velocities (v_i) and n_i is the total number of time steps of the simulation in phase 2. Consequently, $F_{\text{sim},i,j}$ are the contact forces at the corresponding time step at the time t_j and $F_{\text{exp},i,j}$ is the contact force in the experiments, which was linearly interpolated from the experimental data for the time t_j . The factor (t_j/t_{ges}) with $t_{\text{ges}} = 10\text{s}$ was used as weighting factor for a more reliable optimization result.

Optimization of Material Parameter Equations

A mathematical model was constructed to estimate E_0 , E_1 and τ (**Figure 3A**) and the coefficients in these material parameter equations were optimized over the data range specified in **Supplementary Figures 3-6**. To achieve this, 1,920 simulations replicating the AFM mechanical experiments were performed to fine-tune the

coefficients $\alpha_1, \alpha_2, \alpha_3, \beta_1, \gamma_1, \gamma_2, \gamma_3$. Each simulation tested a unique combination of the following variables:

- Elastic modulus (E_0): 500, 1000, 1500, 2000 Pa
- Microgel radius (R): 20, 35, 50 μm
- Maximum contact force (F_1): 2, 4, 6, 8, 10 nN
- Ratio E_1/E_0 : 0.05, 0.1, 0.15, 0.2
- Relaxation time constant (τ): 0.5, 0.1, 0.15, 0.2 s
- Indenter velocity (v): 3 $\mu\text{m/s}$, 5 $\mu\text{m/s}$.

The radius of the indenter cantilever was fixed at 5 μm .

The prediction of the purely elastic modulus (E_0) was based on the relaxed contact force (F_0) and the microgel radius (R). Since only viscoelastic materials were considered, F_0 was not predefined but varied with F_1 , E_1 , τ and v . Error plots were generated for the conditions $E_1/E_0 = 0.05$, $\tau = 0.5$ s, and $v = 3$ $\mu\text{m/s}$, representing the lowest viscosity and largest absolute errors. For each simulation, the values of F_0 , u_{max} , R , F_1 , t_1 and t_{mid} were extracted and used to predict E_0 , τ and E_1 . The relative error for each simulation was calculated as:

$$e_{rel} = \left(\frac{E_{0,sim}}{E_0} \right).$$

The optimization process adjusted the parameters to minimize the maximum relative error across all simulations.

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Conflict of Interest

The authors declare no conflict of interest.

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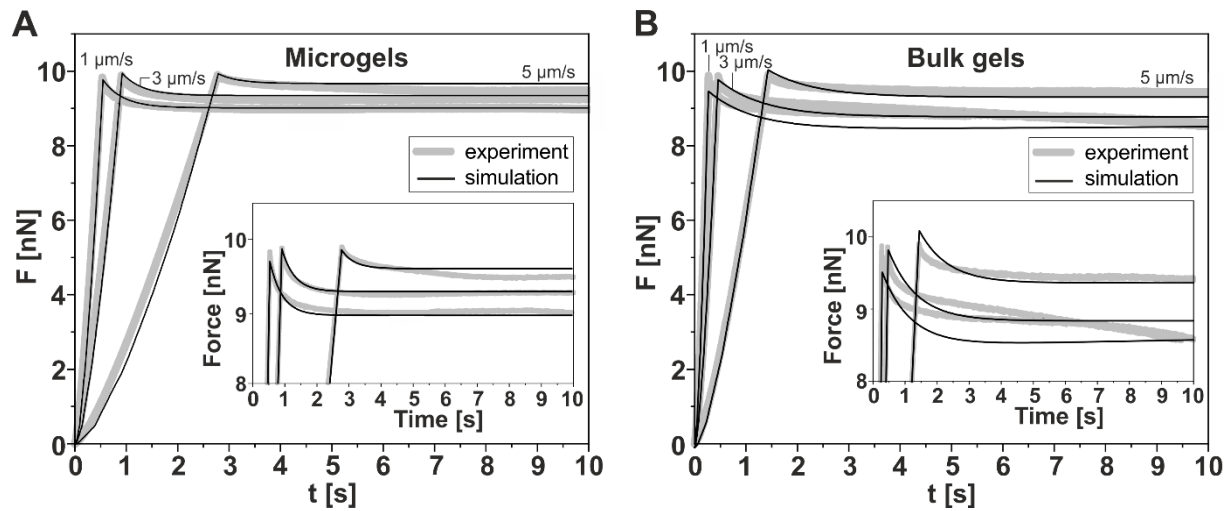
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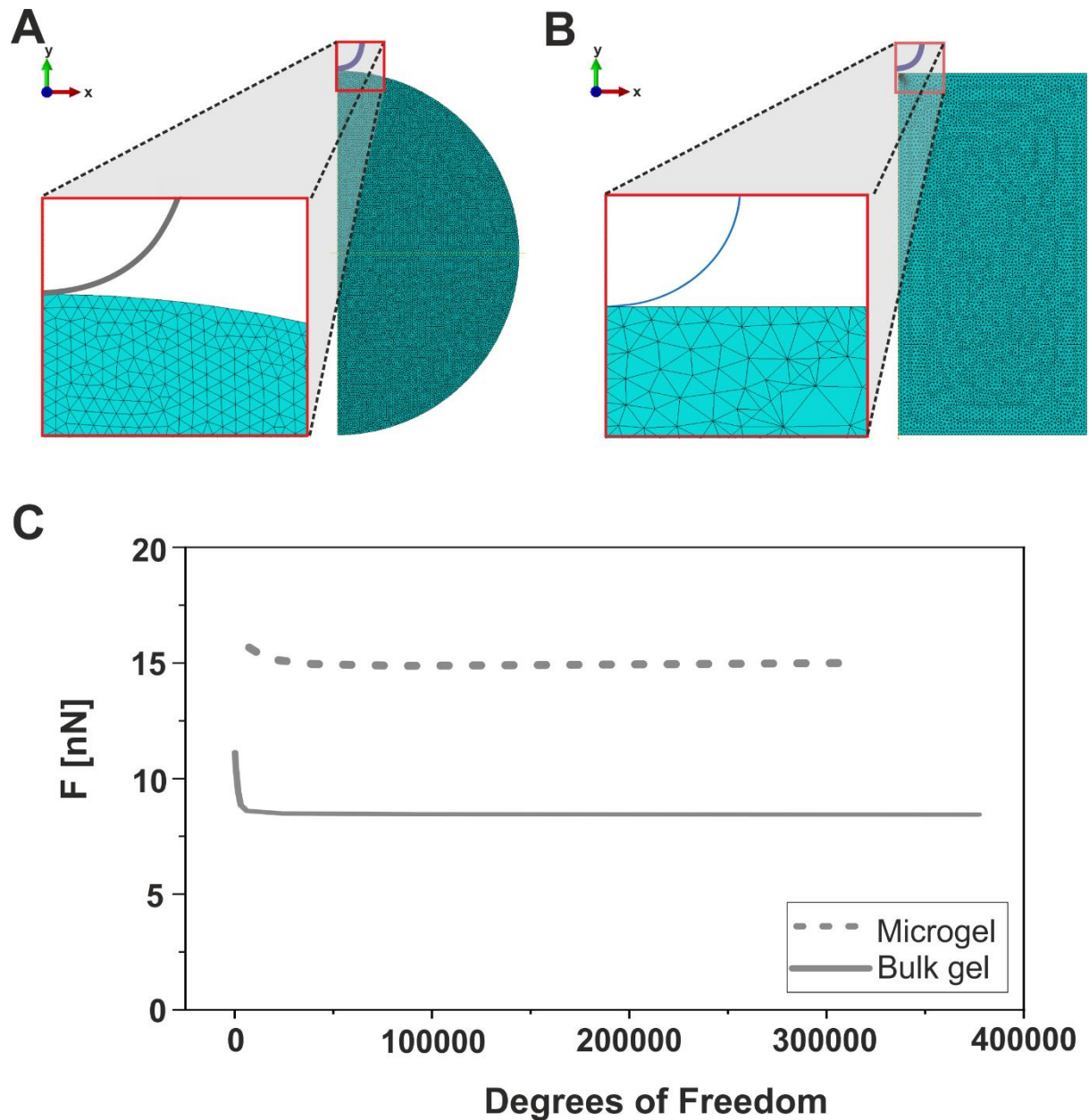
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Supplementary Information



Supplementary Figure 1: Complete stress relaxation phase of microgels and bulk gels. (A) Experimental force–time curves (grey lines) and corresponding FE simulation curves (black lines) for microgels at various indentation velocities, showing the full 10-second holding period of phase 2. (B) Equivalent data for bulk gels, where the force–time curves reveal inconsistent and prolonged relaxation behavior, attributable to poroelastic effects during the holding phase. Insets provide enlarged views of phase 2 for both microgels and bulk gels, highlighting the detailed relaxation dynamics at the probe–sample interface.



Supplementary Figure 2: FE Model Geometry and Mesh Convergence Analysis. FE model geometry and mesh used for nanoindentation simulations of microgels (A) and bulk gels (B). Insets (red boxes) in both panels provide enlarged views of the contact region between the spherical AFM probe and the sample surface, highlighting local mesh refinement. (C) Mesh convergence analysis showing the relationship between computed contact force and increasing degrees of freedom for microgels (dashed line) and bulk gels (solid line), demonstrating that further mesh refinement yields negligible changes in the simulated contact force, thus confirming convergence.

<i>Elastic modulus = 500 Pa</i>					
<i>R = 20 μm</i>		<i>R = 35 μm</i>		<i>R = 50 μm</i>	
<i>F in nN</i>	<i>E₀ in Pa</i>	<i>F in nN</i>	<i>E₀ in Pa</i>	<i>F in nN</i>	<i>E₀ in Pa</i>
2	487.64	2	495.22	2	499.41
4	494.67	4	493.01	4	492.86
6	501.75	6	492.26	6	489.97
8	508.72	8	491.45	8	487.10
10	515.67	10	491.46	10	484.34
<i>Elastic modulus = 1000 Pa</i>					
<i>R = 20 μm</i>		<i>R = 35 μm</i>		<i>R = 50 μm</i>	
<i>F in nN</i>	<i>E₀ in Pa</i>	<i>F in nN</i>	<i>E₀ in Pa</i>	<i>F in nN</i>	<i>E₀ in Pa</i>
2	970.78	2	995.35	2	1013.84
4	975.29	4	990.44	4	998.82
6	982.46	6	988.51	6	990.70
8	989.34	8	986.01	8	985.72
10	996.32	10	984.73	10	982.37
<i>Elastic modulus = 1500 Pa</i>					
<i>R = 20 μm</i>		<i>R = 35 μm</i>		<i>R = 50 μm</i>	
<i>F in nN</i>	<i>E₀ in Pa</i>	<i>F in nN</i>	<i>E₀ in Pa</i>	<i>F in nN</i>	<i>E₀ in Pa</i>
2	1455.52	2	1503.36	2	1521.30
4	1457.24	4	1492.06	4	1511.89
6	1462.93	6	1485.65	6	1498.23
8	1469.84	8	1482.91	8	1487.79
10	1476.99	10	1478.85	10	1485.71
<i>Elastic modulus = 2000 Pa</i>					
<i>R = 20 μm</i>		<i>R = 35 μm</i>		<i>R = 50 μm</i>	
<i>F in nN</i>	<i>E₀ in Pa</i>	<i>F in nN</i>	<i>E₀ in Pa</i>	<i>F in nN</i>	<i>E₀ in Pa</i>
2	1942.10	2	2019.59	2	2058.18
4	1941.57	4	1990.65	4	2027.68
6	1945.68	6	1986.45	6	2012.78
8	1950.58	8	1980.87	8	1997.64
10	1958.36	10	1976.67	10	1989.52

Supplementary Figure 3: Estimated values of the Young's modulus (E_0) compared to actual values, analyzed as a function of the predefined maximum contact force (F) and microgel radius (R).

<i>Tau = 0.5</i>					
<i>R = 20 μm</i>		<i>R = 35 μm</i>		<i>R = 50 μm</i>	
<i>F in nN</i>	<i>tau</i>	<i>F in nN</i>	<i>tau</i>	<i>F in nN</i>	<i>tau</i>
2	0.50628	2	0.50645	2	0.50657
4	0.50593	4	0.50622	4	0.50636
6	0.50571	6	0.50603	6	0.50592
8	0.50544	8	0.50569	8	0.50593
10	0.50531	10	0.50557	10	0.50572
<i>Tau = 1.0</i>					
<i>R = 20 μm</i>		<i>R = 35 μm</i>		<i>R = 50 μm</i>	
<i>F in nN</i>	<i>tau</i>	<i>F in nN</i>	<i>tau</i>	<i>F in nN</i>	<i>tau</i>
2	1.01254	2	1.01071	2	1.01304
4	1.01192	4	1.01235	4	1.01262
6	1.01137	6	1.01201	6	1.01196
8	1.01081	8	1.01124	8	1.01192
10	1.01045	10	1.01097	10	1.01127
<i>Tau = 1.5</i>					
<i>R = 20 μm</i>		<i>R = 35 μm</i>		<i>R = 50 μm</i>	
<i>F in nN</i>	<i>tau</i>	<i>F in nN</i>	<i>tau</i>	<i>F in nN</i>	<i>tau</i>
2	1.51627	2	1.51328	2	1.51704
4	1.51546	4	1.51615	4	1.51622
6	1.51389	6	1.51549	6	1.51556
8	1.51326	8	1.51433	8	1.51469
10	1.51275	10	1.51366	10	1.51413
<i>Tau = 2.0</i>					
<i>R = 20 μm</i>		<i>R = 35 μm</i>		<i>R = 50 μm</i>	
<i>F in nN</i>	<i>tau</i>	<i>F in nN</i>	<i>tau</i>	<i>F in nN</i>	<i>tau</i>
2	2.00582	2	2.00585	2	2.00670
4	2.00582	4	2.00671	4	2.00684
6	2.00154	6	2.00676	6	2.00633
8	2.00056	8	2.00189	8	2.00224
10	1.99956	10	2.00100	10	2.00147

Supplementary Figure 4: Estimated values of the relaxation time (τ) compared to actual values, analyzed as a function of the predefined maximum contact force (F) and the radius (R) of the microgel.

Relaxation modulus = 25 Pa					
$R = 20 \mu\text{m}$		$R = 35 \mu\text{m}$		$R = 50 \mu\text{m}$	
F in nN	E_1 in Pa	F in nN	E_1 in Pa	F in nN	E_1 in Pa
2	24.581	2	24.426	2	24.407
4	24.473	4	24.017	4	23.908
6	24.839	6	24.121	6	23.847
8	25.519	8	24.454	8	24.072
10	26.361	10	25.014	10	24.459
Relaxation modulus = 50 Pa					
$R = 20 \mu\text{m}$		$R = 35 \mu\text{m}$		$R = 50 \mu\text{m}$	
F in nN	E_1 in Pa	F in nN	E_1 in Pa	F in nN	E_1 in Pa
2	48.152	2	47.895	2	47.832
4	48.174	4	47.361	4	47.081
6	49.086	6	47.661	6	47.158
8	50.472	8	48.357	8	47.688
10	52.191	10	49.466	10	48.475
Relaxation modulus = 75 Pa					
$R = 20 \mu\text{m}$		$R = 35 \mu\text{m}$		$R = 50 \mu\text{m}$	
F in nN	E_1 in Pa	F in nN	E_1 in Pa	F in nN	E_1 in Pa
2	71.903	2	71.361	2	71.344
4	72.320	4	71.065	4	70.477
6	73.825	6	71.763	6	70.925
8	75.974	8	72.958	8	71.876
10	78.673	10	74.674	10	73.209
Relaxation modulus = 100 Pa					
$R = 20 \mu\text{m}$		$R = 35 \mu\text{m}$		$R = 50 \mu\text{m}$	
F in nN	E_1 in Pa	F in nN	E_1 in Pa	F in nN	E_1 in Pa
2	96.089	2	95.366	2	94.796
4	97.011	4	95.472	4	94.693
6	99.224	6	96.595	6	95.558
8	102.362	8	98.509	8	97.052
10	106.030	10	100.979	10	98.879

Supplementary Figure 5: Estimated values for the relaxation modulus (E_1) at the lower boundary (with $E_0 = 500$ Pa) compared to actual values, analyzed as a function of the predefined maximum contact force (F) and the radius (R) of the microgel.

Relaxation modulus = 100 Pa					
$R = 20 \mu\text{m}$		$R = 35 \mu\text{m}$		$R = 50 \mu\text{m}$	
F in nN	E_1 in Pa	F in nN	E_1 in Pa	F in nN	E_1 in Pa
2	101.846	2	96.8616	2	101.809
4	100.060	4	96.8616	4	99.651
6	99.015	6	96.8616	6	98.332
8	98.312	8	96.8616	8	97.630
10	97.875	10	96.8616	10	96.683
Relaxation modulus = 200 Pa					
$R = 20 \mu\text{m}$		$R = 35 \mu\text{m}$		$R = 50 \mu\text{m}$	
F in nN	E_1 in Pa	F in nN	E_1 in Pa	F in nN	E_1 in Pa
2	197.908	2	198.019	2	198.029
4	195.455	4	194.523	4	194.558
6	193.773	6	192.965	6	192.512
8	192.613	8	191.578	8	191.324
10	192.321	10	190.113	10	189.870
Relaxation modulus = 300 Pa					
$R = 20 \mu\text{m}$		$R = 35 \mu\text{m}$		$R = 50 \mu\text{m}$	
F in nN	E_1 in Pa	F in nN	E_1 in Pa	F in nN	E_1 in Pa
2	293.290	2	293.237	2	292.713
4	290.422	4	288.993	4	289.209
6	288.803	6	286.562	6	286.938
8	287.617	8	285.443	8	285.376
10	287.124	10	284.614	10	283.782
Relaxation modulus = 400 Pa					
$R = 20 \mu\text{m}$		$R = 35 \mu\text{m}$		$R = 50 \mu\text{m}$	
F in nN	E_1 in Pa	F in nN	E_1 in Pa	F in nN	E_1 in Pa
2	388.640	2	387.685	2	387.658
4	385.592	4	384.285	4	383.811
6	384.446	6	382.171	6	382.422
8	384.358	8	381.458	8	379.188
10	384.400	10	380.968	10	379.736

Supplementary Figure 6: Estimated values for the relaxation modulus (E_1) at the upper boundary (with $E_0 = 2000$ Pa) compared to actual values, analyzed as a function of the predefined maximum contact force (F) and the radius (R) of the microgel.